

Advancing battery research through large language models: A review

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GRAPHICAL ABSTRACT



PUBLIC SUMMARY

- Large Language Models (LLMs) enable a paradigm shift in battery research, addressing multilevel innovation needs.
- LLMs accelerate iteration across battery knowledge, materials development, manufacturing, and system management.
- This work examines the potential of LLMs in battery research, addressing current challenges and future directions.

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Rechargeable batteries are pivotal for achieving carbon neutrality and enabling the renewable energy transition. Their advancement requires innovations at micro (materials), device (manufacturing), and system (control and optimization) levels. However, traditional trial-and-error approaches are inadequate for modern scientific demands. As a transformative artificial intelligence (AI) technology, large language models (LLMs) deliver powerful semantic understanding and reasoning capabilities, driving a paradigm shift in battery research to address multilevel innovation needs. Nevertheless, this field still faces dual challenges: ambiguous technical roadmaps and fragmented progress in stage-specific achievements. This review systematically consolidates recent advances in applying LLMs to battery research, distilling core findings across four critical domains: knowledge integration, materials discovery, manufacturing processes, and system management. To address key bottlenecks—including limited model interpretability, inadequate alignment with electrochemical mechanisms, and real-world data adaptation challenges—we propose structured frameworks for deep integration of battery research and LLMs, alongside defined future technical pathways. These frameworks bridge fundamental battery science with AI-driven innovation paradigms to facilitate groundbreaking advances in next-generation battery technologies.

INTRODUCTION

With the growing global demand for sustainable green energy, rechargeable batteries have become essential components in a wide range of applications, including electric vehicles¹ and power grids.² According to the Web of Science, battery-related publications have surpassed 1.93 million as of 2024, with a significant surge in research activities especially in the last decade. However, the explosive growth of literature has overwhelmed traditional methods of manual reading and analysis, making it increasingly difficult to stay current with recent advances or systematically retrieve and interpret past research findings. Consequently, a vast amount of valuable scientific knowledge and data remains fragmented and underutilized across the literature landscape. Therefore, the scientific community is in urgent need of tools that can effectively distill and synthesize vast amounts of information into actionable insights.

The development of large language models (LLMs) represents a transformative advancement in artificial intelligence (AI) technology, offering powerful capabilities in semantic understanding, reasoning, and summarization.^{3–5} A major breakthrough came in 2018 with the release of bidirectional encoder representations from transformers (BERT) by Google, which demonstrated impressive contextual comprehension abilities.⁶ This progress was further accelerated by the introduction of ChatGPT by OpenAI in 2022, marking a new era for LLMs.^{7–10} ChatGPT, in particular, exhibits advanced skills in text interpretation and summarization, enabling in-depth analysis of text and efficient extraction of critical insights. The interactive question-and-answer (Q&A) format of GPT has further lowered the barriers to AI technology, reshaping the paradigm of human-computer collaboration. In early 2025, DeepSeek's open-source architecture reduced the cost and technical barriers of LLM applications, democratizing the accessibility of AI technology in a cost-effective manner.¹¹ The rise of the "AI for Science" (AI4S) paradigm has driven a deep integration of AI technologies with scientific discovery.¹² Leveraging their exceptional knowledge comprehension and logical reasoning capabilities,

LLMs provide researchers with intelligent tools that efficiently distill and synthesize vast amounts of information into actionable insights, thereby accelerating the pace of scientific and technological advancement. These models now support breakthroughs in data analysis, pattern recognition, literature mining, and even experimental planning. Their applications span across disciplines, including physics,¹³ biology,¹⁴ chemistry,¹⁵ and materials science,¹⁶ fostering interdisciplinary innovation and driving the emergence of new scientific theories and methodologies.

In the era of AI4S, battery researchers are actively exploring the integration of LLMs into their workflows.^{17–19} On one hand, LLMs are being utilized to extract critical insights from the rapidly expanding corpus of battery science literature,^{20,21} enabling researchers to quickly identify emerging research frontiers and discover novel materials with tailored performance attributes. On the other hand, LLMs are also being employed to optimize battery manufacturing processes and monitor system performance, thereby improving product quality, reliability, and lifespan.²²

This review presents a comprehensive overview of recent advances and applications of LLMs in battery research, focusing on four key areas: battery knowledge integration, materials development, intelligent manufacturing, and battery system management, as shown in Figure 1. Given the emerging nature of LLM applications in this field, the review includes both peer-reviewed publications and preprints.

BATTERY KNOWLEDGE INTEGRATION

As shown in Figure 2A, we conducted a statistical analysis of battery-related scientific publications over the past decade using data from the Web of Science. The results reveal a consistent upward trend in publication volume, with the number of battery-related papers in 2024 reaching approximately 2.7 times that of 2014. Notably, key subfields such as sodium-ion batteries, solid-state batteries, and lithium-ion batteries have experienced particularly rapid growth. As illustrated in Figure 2B, the number of publications in 2024 related to sodium-ion batteries, solid-state batteries, and lithium-ion batteries increased by approximately 9.2-, 5-, and 2.5-fold, respectively, compared with 2014. It is worth noting that, over the past decade, literature containing the keywords "battery," "sodium-ion battery," "solid-state battery," and "lithium-ion battery" accounted for 73.7%, 92.4%, 79.4%, and 79.8% of total retrieved records in their respective queries. This dramatic surge in scientific literature reflects not only the escalating global interest in battery technologies but also underscores the increasing difficulty of relying on traditional manual-intensive methods for literature review and knowledge extraction.

As the volume of publications continues to grow, there is an urgent need for more intelligent and efficient tools of AI to explore the vast scientific corpus and support comprehensive knowledge integration. By leveraging advanced AI technologies, scientists can efficiently extract key insights from vast amounts of literature, establish or expand specialized databases, and identify core challenges and emerging trends across research fields.^{23,24} This, in turn, can significantly expedite the advancement and innovation of battery technologies.

In the early stages, researchers developed rule-based toolkits to extract information from battery-related texts.^{25–27} While effective in specific contexts, these approaches required extensive expert input to define extraction rules, limiting their scalability and adaptability. With the development of machine learning, algorithms such as neural networks were introduced to improve both the efficiency and accuracy of battery information mining.^{28–33} However,

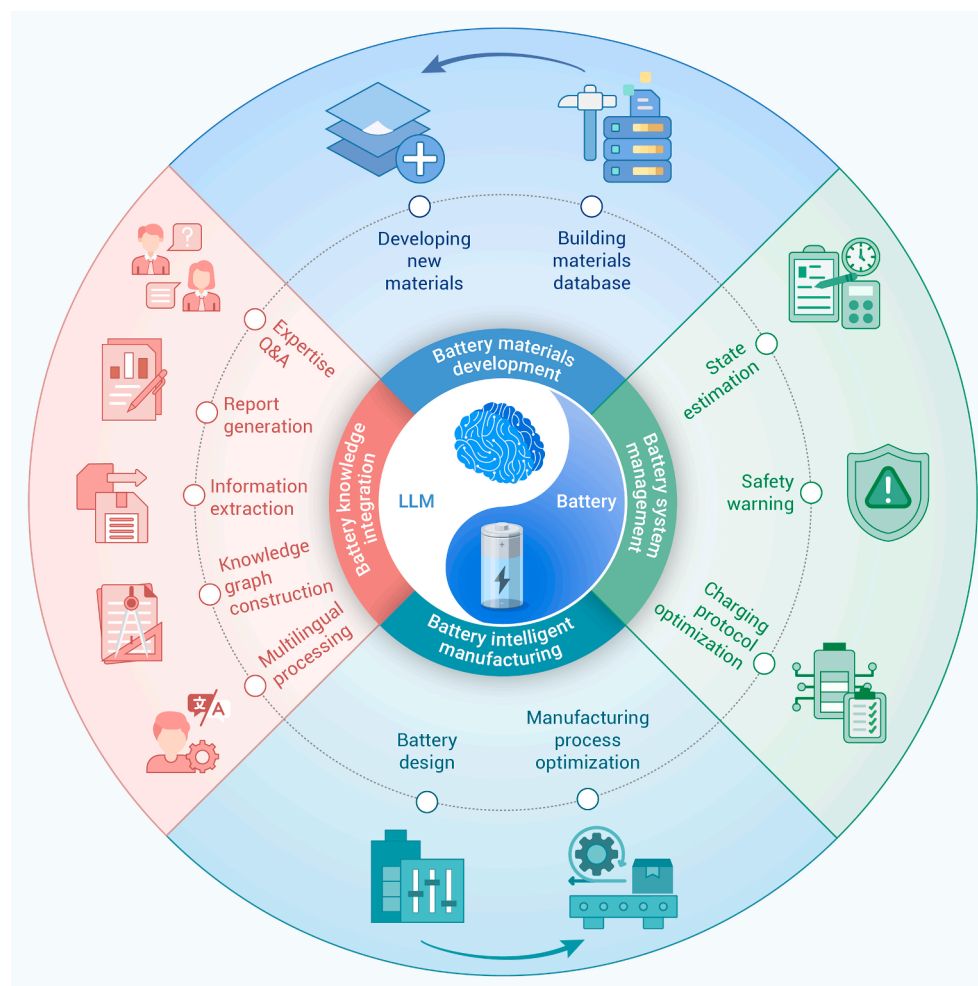


Figure 1. The emerging roles of LLMs in battery science and engineering

of knowledge graphs. Through these applications, LLMs enable efficient knowledge integration and enhance research productivity in the battery domain. Among the three layers in the development pipeline of LLM in knowledge integration, the model layer represents the most critical component.³⁷

Regarding the first strategy in the model layer, expert LLMs have demonstrated impressive capabilities in fields such as biology,³⁸ chemistry,¹⁵ and materials science,^{39,40} particularly in tasks like information extraction and domain-specific Q&A. In the battery field, researchers have recently developed expert LLM models such as BatteryBERT, designed to mine key information on cathode, anode, and electrolyte materials from scientific papers.⁴¹ Other expert LLMs have also been proposed for extracting manufacturing-related data from broad literature.⁴² However, the development of expert LLMs presents significant challenges. It requires access to large-scale, high-quality, domain-specific annotated corpora, as well as substantial computational resources for model training and fine-tuning. These resource-intensive demands make expert LLM development both complex and costly, potentially limiting their widespread adoption in the near term.

As for the second strategy in the model layer, prompt engineering typically utilizes a multi-turn Q&A approach, enabling general-purpose LLMs to incrementally build understanding through a

these models often need task-specific training, which hinders their generalizability.

Recently, the rapid development of LLMs has brought transformative improvements in this knowledge integration area. Equipped with powerful contextual understanding and reasoning capabilities, LLMs—especially when paired with vectorization techniques—can read and comprehend entire scientific papers much like human experts. Through interactive Q&A formats, users can extract essential information directly from the scientific corpus.³⁴ In addition, LLMs support multilingual processing and natural text generation, enabling communication with researchers across linguistic boundaries.³⁵ Their ability to synthesize complex knowledge also facilitates the automated generation of research reports and construction of knowledge graphs.³⁶

As shown in Figure 3, the development pipeline for applying LLMs to scientific knowledge integration of battery literature can be mainly divided into three layers: the data layer, model layer, and task layer.³⁴ The data layer focuses on collecting battery-related content from diverse sources, including research papers, patents, books, and other technical documents. This stage also involves standardizing and preprocessing data stored in various formats (e.g., PDF, XML, HTML, TXT) and expression types (e.g., text, images, tables, and formulas), ensuring consistency and usability for downstream modeling. In the model layer, the first step is to select an appropriate LLM base model. Then researchers can then choose between three common strategies: (1) fine-tuning the base model on domain-specific data to develop a specialized battery-focused LLM, (2) leveraging prompt engineering to guide a general LLM without additional training, or (3) building a specialized knowledge vector database and adopting retrieval-augmented generation (RAG) technology enhance domain-specific capabilities of general models while mitigating model hallucinations. This layer also includes model testing and iterative optimization to improve performance and reliability. The task layer involves applying the trained or guided LLM to specific downstream tasks. These may include Q&A, automated report generation, information extraction, multilingual translation, and the construction

sequence of carefully crafted prompts.^{43–45} This strategy enhances precision and efficiency in various tasks, including domain-specific information extraction, expert-level Q&A, and knowledge graph construction. However, manually designed prompts face three inherent limitations: time-consuming nature, subjectivity, and domain dependency. The design process requires substantial human resources and is susceptible to researchers' cognitive biases; developing prompts for specialized data such as chemical formulas and molecular structures demands deep involvement from domain experts, significantly raising cross-disciplinary application barriers. In contrast, automated prompt generation techniques (e.g., methods based on optimization algorithms and feedback iteration) emerge as promising alternatives through algorithmic approaches: substantially reducing manual intervention to lower trial-and-error costs, replacing subjective judgments with quantitative metrics to enhance prompt consistency and scalability, while leveraging batch-generated domain-adaptive prompt templates to overcome specialized knowledge limitations.^{46,47} These automated techniques can improve LLMs' task comprehension capabilities and better unleash their potential. However, it must be emphasized that both manual and automated prompt engineering are ultimately constrained by the fundamental ceiling of the underlying LLM's domain comprehension capability. When models lack specialized knowledge (e.g., in predicting multi-scale material behaviors in battery science), even optimal prompt strategies may fail.⁴⁰

Two common issues associated with LLMs are knowledge cutoff and hallucination. Knowledge cutoff refers to the fact that the training data of an LLM does not include content beyond its training cutoff date, meaning that its knowledge is limited to the data available during training. Hallucination, on the other hand, occurs when a user's query exceeds the scope of the LLM's knowledge. In such cases, the model generates responses based on its own understanding, which may be inaccurate. The advent of RAG has addressed these knowledge limitations by integrating external data. Essentially, RAG endows LLMs with a real-time external brain (knowledge base), thereby enhancing the accuracy,

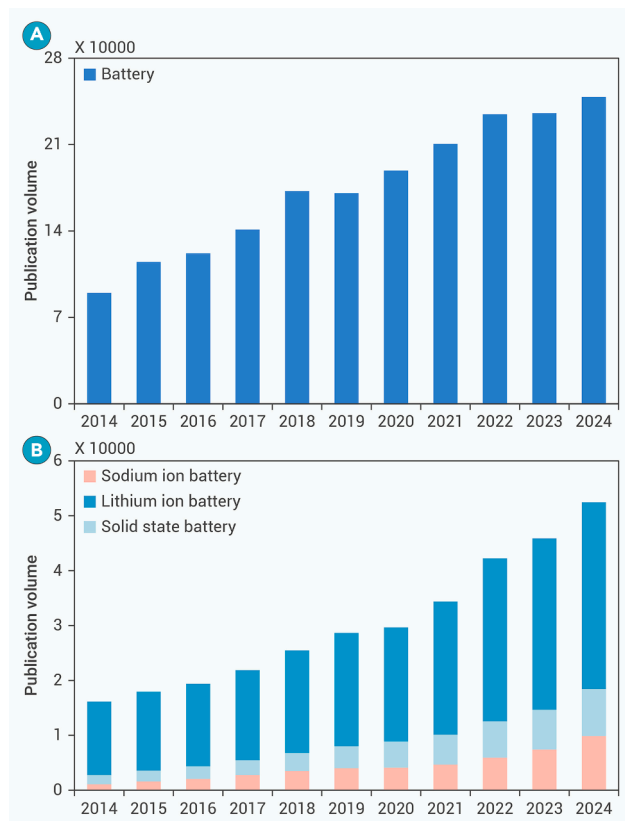


Figure 2. Statistical analysis of battery-related publications (A) The number of papers retrieved with “battery” as the search term in Web of Science from 2014 to 2024. (B) The number of papers retrieved with “sodium-ion battery,” “solid-state battery,” “lithium-ion battery” as the search terms in Web of Science from 2014 to 2024.

reliability, and timeliness of their responses while maintaining relatively low implementation and maintenance costs.⁴⁸ It dynamically retrieves relevant information from an external knowledge base and leverages this information to guide LLMs in generating more accurate, reliable, and contextually appropriate responses. This effectively addresses the knowledge limitations and hallucination issues inherent in traditional large models. RAG can be used in conjunction with both fine-tuned LLMs and general-purpose models. For instance, Zhao et al.⁴⁹ developed a lithium-ion solid-state assistant named LISA based on the RAG architecture. This assistant serves as a research aid for knowledge fragment organization in battery science and other fields. Specifically tailored for the solid-state battery research domain, LISA enhances information retrieval and extraction. This system provides a powerful open-source framework for the future development of automated knowledge retrieval, understanding, and management, particularly in supporting the development of new materials. Meanwhile, Liu et al.⁵⁰ systematically utilized an integrated RAG workflow aimed at automating and simplifying research in battery recycling. The system was validated in the context of battery recycling and demonstrated excellent performance in literature review tasks. Leng et al.⁵¹ developed ChatSSB, a research assistant for solid-state battery materials based on LLM technology, as shown in Figure 4. Leveraging the RAG framework, ChatSSB excels in extracting precise information from the latest research publications through an intuitive Q&A interface. Beyond its foundational capabilities, ChatSSB boasts a customizable expert knowledge database, continuously updated through a dynamic feedback mechanism. This ensures researchers access to cutting-edge and reliable information, overcoming the limitations of outdated or incomplete literature.

BATTERY MATERIALS DEVELOPMENT

The development of battery materials requires comprehensive consideration of a wide range of chemical and physical properties, such as redox potentials, chemical stability, crystal structures, mechanical strength, or conductivity. Reliable materials design also depends on detailed data during processes such as material synthesis, characterization, or performance testing.^{52–58} While re-

searchers have curated portions of these data into structured databases, a vast amount remains embedded in unstructured formats within scientific literature and patents. Effectively organizing and structuring this multi-source, heterogeneous data is a crucial step toward advancing battery materials research.⁵⁹ By transforming this unstructured information into structured databases, researchers can significantly improve materials screening efficiency and performance prediction accuracy, thereby accelerating the discovery of novel materials and reducing experimental costs.⁶⁰ Despite these efforts, the pace of innovation in battery materials continues to lag behind the rapidly growing demands of industry. Human-driven approaches alone are no longer sufficient. Deeper integration of AI, particularly LLMs, is essential. In Miret and Krishnan,⁶¹ the authors outlined a Materials Science LLM that is based on domain knowledge and capable of generating hypotheses followed by hypothesis testing. They also discussed the application of Materials Science LLM to the six interrelated steps of real-world material discovery: material querying, data retrieval, material design, internal evaluation, experimental planning, and experimental execution. This provides new ideas for intelligent, automated new material development based on LLM.

Based on existing research, this paper summarizes a framework for developing battery materials based on LLMs, as shown in Figure 5. LLMs play an integral role in the end-to-end workflow of battery materials development, spanning from information extraction and hypothesis generation to downstream task execution. In Figure 5A, LLMs—guided by prompt-based inputs—extract structured information from diverse data sources such as scientific papers, patents, books, and experimental records. The resulting battery material databases contain rich, multi-dimensional data including material-named entity, chemical composition, microstructure information, physical property, and synthesis method, providing a solid foundation for materials research. Here, the general LLMs can be guided to complete specific tasks through prompt engineering, or an expert LLM can be developed for specific tasks through fine-tuning. Building on this, Figure 5B illustrates the core reasoning process enabled by expert LLMs. Researchers interact with the expert LLM through a multilayered Q&A process, posing questions related to boundary conditions, key performance indicators, and application scenarios. Based on this input, the expert LLM generates research hypotheses, such as material development program, feasibility analysis, and which experimental equipment is needed. These hypotheses are then passed to a downstream task interface for further processing. In the downstream task phase, as shown in Figure 5C, researchers continue to interact with the expert LLM to perform specific functions such as potential material screening, property prediction, and synthesis path optimization. Ultimately, this extends to automated material preparation, characterization, and testing. Through this iterative and interactive workflow, expert LLMs enable a closed-loop integration of data-driven insights, hypothesis formulation, and experimental execution—offering a transformative approach to accelerate battery materials innovation.

Battery materials database

A comprehensive materials database is essential for the advancement of battery materials research, as it consolidates large volumes of experimental results, computational data, and scientific literature. By analyzing these data, researchers can uncover correlations between material structures and properties, predict the performance of novel materials, and optimize existing formulations. Such databases also support the design of more efficient experimental protocols, reducing trial-and-error efforts and improving the success rate of experimental validation. However, traditional manual methods for database construction are time-consuming, labor-intensive, and susceptible to human error. These methods often struggle with data consistency and completeness, and are particularly inefficient at processing unstructured data from diverse sources. As a result, the update frequency and practical utility of traditional databases are limited, which in turn constrains research efficiency and the pace of innovation.

In contrast, the application of LLMs can significantly improve the efficiency and scalability of battery materials database construction. LLMs can automatically extract key information from vast quantities of literature and experimental records, integrating and standardizing heterogeneous data to accelerate database generation and maintenance.⁶² For instance, Huang et al.^{41,63} developed BatteryDataExtractor, a literature mining toolkit for battery materials that uses BERT-based models for token classification. As shown in Figure 6A, the

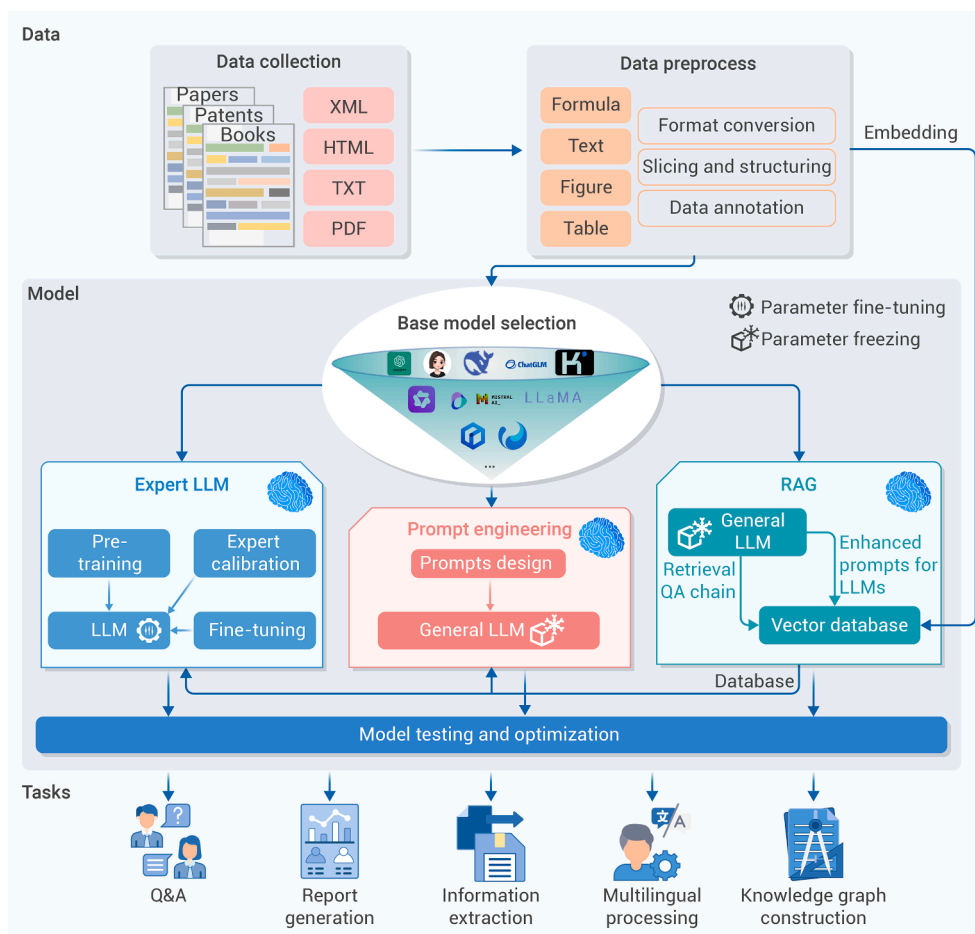


Figure 3. The development pipeline for applying LLMs to scientific knowledge integration of battery literature

cated HEA arrays with diverse element compositions. Scanning microscopy was then employed for high-throughput characterization, successfully identifying HEA compositions with high catalytic activity. In another study, Yu et al.⁷³ leveraged LLMs along with extensive domain knowledge to perform material screening and evaluation through self-retrieval and self-assessment mechanisms. Jia et al.¹⁶ proposed a novel material design framework driven by LLM agents, where human instructions are translated into actionable material modifications. This framework supports interpretable and goal-driven material discovery, marking a significant step toward fully autonomous materials research. Robson et al.⁷⁴ proposed a multi-agent network based on gpt-4-0125-preview that successfully generated aqueous deep eutectic electrolyte compositions for zinc-ion batteries. By analyzing academic papers from relevant fields, the network identified innovative, cost-effective, and sustainable Lewis base candidates. Chen et al.⁷⁵ proposed a chained LLM-human interaction framework that employs GPT-4 for specialized research in chemistry and materials science, focusing on identifying electrolyte materials for the emerging four-electron zinc-iodine (Zn-I2) battery. Through this interactive workflow, GPT-4 re-

trieves key literature information and recommends promising electrolyte materials for experimental evaluation, thereby elucidating the mechanisms of multi-electron zinc-iodine batteries, identifying suitable materials, and enhancing the performance of four-electron Zn-I2 batteries.

LLM-based approaches for predicting material performance are gaining increasing attention. Leveraging structured battery material databases, LLMs can assess the physical and chemical properties of candidate materials.⁷⁶ By integrating LLMs with deep learning algorithms such as graph neural networks, the accuracy of material performance prediction can be enhanced.⁷⁷ Researchers have applied LLM techniques to predict the properties of various battery components, such as cathode materials, anode materials, liquid electrolytes, and solid-state electrolytes. For instance, Lei et al.⁷⁸ constructed a comprehensive knowledge graph encompassing material structures, properties, and research methodologies. By integrating this knowledge graph with GPT-4, they developed a joint reasoning approach for capacity performance prediction of anode and cathode materials for batteries. Soares et al.⁷⁹ introduced a multimodal framework based on the chemical language model MolFormer, developed to explore the design space of liquid electrolytes. This approach leverages chemical information pre-trained on large unlabeled corpora and demonstrates superior performance in electrolyte Coulombic Efficiencies property prediction.

Developing new materials

In the field of materials development, AI systems based on LLMs have demonstrated remarkable capabilities in predicting material properties, forecasting the outcomes of chemical reactions, and autonomously designing, planning, and executing complex experiments.^{68–71} This enables end-to-end intelligent automation across the entire materials development pipeline—from structural design and performance prediction to synthesis optimization, material preparation, characterization, and testing.

For material screening, LLMs enhance efficiency by enabling natural language interaction and integrating with high-throughput experimental techniques to rapidly identify promising material structures or compounds. For example, Pan et al.⁷² introduced a high-throughput screening method for efficient electrocatalysts based on Pt-based high-entropy alloys (HEAs). By utilizing an element library generated by LLMs and combining it with micro-scale precursor printing and pulsed high-temperature synthesis technologies, they fabri-

cates HEA arrays with diverse element compositions. Scanning microscopy was then employed for high-throughput characterization, successfully identifying HEA compositions with high catalytic activity. In another study, Yu et al.⁷³ leveraged LLMs along with extensive domain knowledge to perform material screening and evaluation through self-retrieval and self-assessment mechanisms. Jia et al.¹⁶ proposed a novel material design framework driven by LLM agents, where human instructions are translated into actionable material modifications. This framework supports interpretable and goal-driven material discovery, marking a significant step toward fully autonomous materials research. Robson et al.⁷⁴ proposed a multi-agent network based on gpt-4-0125-preview that successfully generated aqueous deep eutectic electrolyte compositions for zinc-ion batteries. By analyzing academic papers from relevant fields, the network identified innovative, cost-effective, and sustainable Lewis base candidates. Chen et al.⁷⁵ proposed a chained LLM-human interaction framework that employs GPT-4 for specialized research in chemistry and materials science, focusing on identifying electrolyte materials for the emerging four-electron zinc-iodine (Zn-I2) battery. Through this interactive workflow, GPT-4 re-

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In synthetic pathway optimization, LLMs can propose innovative reaction routes and conditions. For example, Reinhart and Statt⁸⁰ demonstrated that the Claude 3.5 model outperforms active learning schemes in screening monomer sequences to achieve targeted morphologies in macromolecular self-assembly. Utilizing pre-trained language models significantly reduces the need for hyperparameter tuning while introducing novel capabilities like self-reflection. Lu et al.⁸¹ integrated LLMs into an evolutionary optimization framework for single- and multi-objective optimization of functional transition metal complexes. The study revealed that LLMs, as generative models, can recommend novel ligands and transition metal complexes with distinctive properties by integrating domain-specific knowledge with external chemical data, thus combining the advantages of efficient optimization and molecular generation.

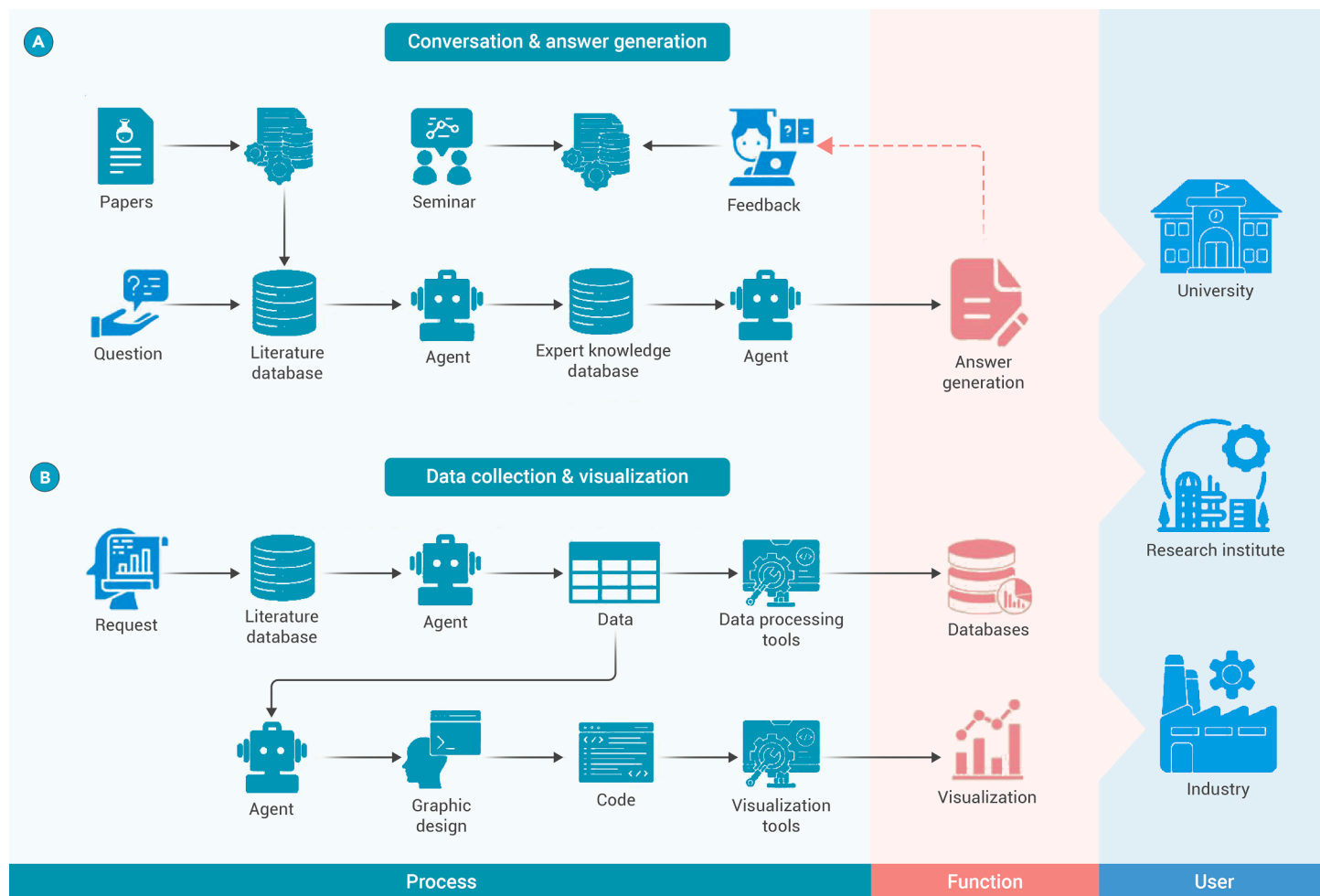


Figure 4. ChatSSB workflow: reducing hallucinations with RAG and streamlining research data collection and visualization (A) RAG architecture for reducing hallucinations in ChatSSB. (B) Data collection and visualization workflow in ChatSSB. Reproduced with permission.⁵¹ Copyright 2025, American Chemical Society.

In material preparation, LLMs are capable of generating detailed experimental protocols and operational steps. When combined with intelligent machinery or robots, they can guide automated synthesis processes. For instance, Ni et al.⁸² implemented natural language interaction with a multi-agent system to enable autonomous control of material preparation, significantly enhancing experimental precision and reproducibility.

LLMs also enhance the efficiency of materials characterization and testing by integrating multimodal data—including electron microscopy techniques (e.g., scanning electron microscopes,⁸³ transmission electron microscopes⁸⁴) and analytical methods such as spectroscopy (e.g., Raman spectroscopy⁸⁵) and diffraction (e.g., X-ray diffraction⁸⁶). Leveraging transformer-based multi-head attention mechanisms, LLMs can simultaneously analyze lattice distortions, phase interfaces, and elemental distributions.^{87–89} This enables the construction of quantitative models linking ion transport pathways to macroscopic properties like electrical conductivity, thereby accelerating structure-property relationship analysis. Moreover, LLM-powered code generation is revolutionizing equipment control. The Febba team, for example, used ChatGPT-4 to develop control software for materials characterization equipment, simplifying setup and execution while allowing researchers to focus on data analysis.⁹⁰ Additionally, multimodal LLMs enhanced by vision-language pre-training can recommend relevant literature based on uploaded microscopic images or spectroscopic data.⁹¹

A prominent example of full-process automation is the A-Lab platform developed by Professor Gerbrand Ceder's team (Figure 6C).⁶⁵ This system uses LLMs to extract synthesis formulas from literature and databases, applies active learning algorithms based on thermodynamic models to refine them, and employs robotic systems to perform synthesis, characterization, and quality evaluation. Compared with traditional manual workflows, this LLM-driven approach improves materials development efficiency by an order of magnitude

and significantly reduces R&D costs.⁶⁰ Zheng et al.⁶⁶ discussed how to customize “chemically aware” models for specific tasks and integrate them into existing mesh-like chemical practices, transforming traditional “make, characterize, use” protocols driven by empirical knowledge into discovery cycles based on identifying synthesis-structure-property relationships. Additionally, they explored integrating modular LLM agents into multi-agent laboratory systems (e.g., autonomous robotic labs) to streamline labor-intensive tasks and collaborate with chemists, as well as how LLMs can reduce barriers to applying generative AI and data-driven workflows to challenging research problems such as crystallization, as shown in Figure 6D.

BATTERY INTELLIGENT MANUFACTURING

Battery manufacturing research encompasses both battery design and production. Effective battery design requires careful consideration of key performance requirements such as cost, application scenarios, geometric requirements, durability, and energy density.^{92–94} Meanwhile, the cell manufacturing process involves complex steps including slurry mixing, electrode coating-drying-calendering-slitting-stacking, battery assembly, electrolyte filling, battery formation and aging, each of which significantly influences the final product's performance and reliability.⁹⁵ Currently, optimization of design and manufacturing processes largely depends on trial-and-error experimentation and expert-driven heuristics. These conventional approaches are costly, time-consuming, and often lack the precision needed for high-quality, scalable production. In this context, LLMs offer promising AI-driven strategies to accelerate and enhance battery manufacturing.

As illustrated in Figure 7, expert LLMs play a central role in enabling intelligent, end-to-end battery manufacturing by bridging human design input, machine execution, and quality feedback. In Figure 7A, an expert battery manufacturing LLM is developed by training on diverse manufacturing-related data sources,

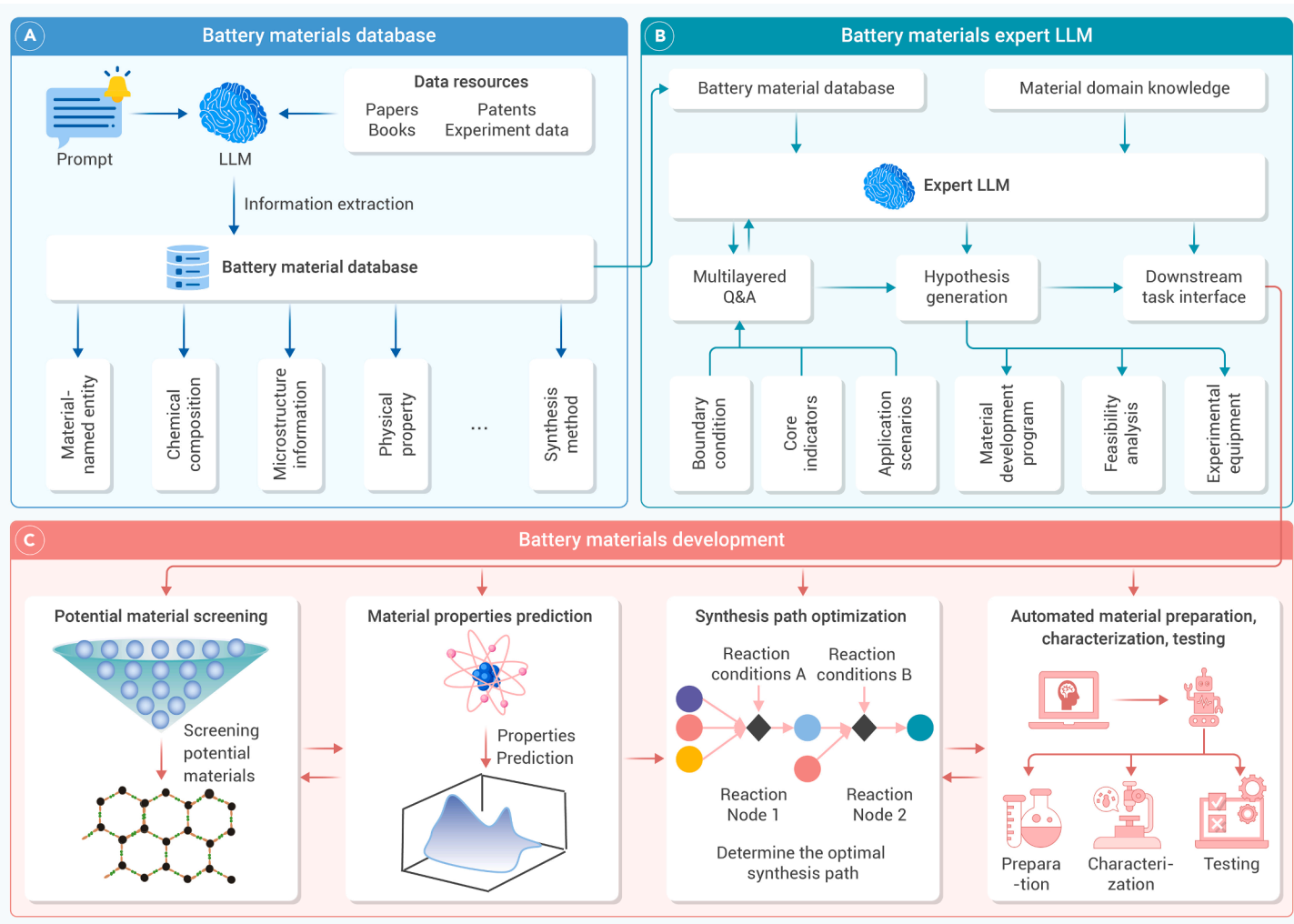


Figure 5. LLM-enabled workflow for battery materials development (A) Data extraction and structuring from multi-source literature using LLMs. (B) Hypothesis generation through multi-turn Q&A between researchers and LLMs. (C) LLM-supported downstream applications including screening, prediction, optimization, and automated characterization and testing.

including literature and patent databases, battery materials and design datasets, production process documentation, and performance testing results. This expert model accumulates comprehensive knowledge across the entire manufacturing workflow. In Figure 7B, battery design engineers input performance requirements—such as energy density, geometric constraints, thermal protection, cost, and durability—into the expert LLM. The model then generates detailed design parameters at different levels, including material parameters such as specific capacity, electrode parameters such as areal loading, and battery parameters such as nominal voltage. Through conversational Q&A interactions, LLMs can provide battery manufacturers with reference solutions or computational models for optimizing product performances.⁹⁶ This interaction also exemplifies effective human-AI collaboration in the battery design, where human oversight can still verify and adjust the results if necessary. Moving to Figure 7C, utilizing the generated and proofread battery design parameters, the LLM can then generate corresponding material ratios, electrode manufacturing processes, assembly procedures, environmental requirements, equipment specifications, and performance testing parameters for battery manufacturing. Finally, for some automated equipment, the LLM can generate machine-executable scripts based on the process parameters to achieve automated material preparation.⁹⁷ These are automatically delivered to each stage of the production line—including mixing, coating, drying, calendaring, slitting, stacking, assembly, electrolyte filling, formation, and aging.^{98,99}

In battery design, LLMs can provide technical services in two aspects: knowledge graph-driven parameter optimization and multimodal collaborative design. Although LLMs cannot directly perform physical design or draw engineering diagrams, they can integrate global battery research papers, patent databases,

and corporate historical data to construct a multimodal electrochemical knowledge graph.¹⁰⁰ Based on the causal reasoning capabilities of LLMs, electrode material selection and structural design schemes that meet specific requirements such as energy density and fast-charging rates can be automatically generated. In addition, LLM-based knowledge Q&A systems can provide informational support and simulation-assisted decision-making. Leveraging the strengths of multimodal large models, industrial personnel can interact with the LLM assistant through voice recognition, image processing, and even augmented reality scenarios.^{96,101,102} This human-machine interaction mode can improve design efficiency and reduce costs.

The involvement of LLMs in battery manufacturing can enhance automation levels, optimize production processes, and improve product quality. Li et al.¹⁰³ integrated the battery vacuum drying physical model into AgentGPT via an application programming interface (API), as shown in Figure 8A. The LLM then decomposes the user's question, creates a task set, and proactively activates this model for detailed calculations. It subsequently provides the user with corresponding answers and information. The collaborative system between the LLM and the physical model not only enhances the simulation capabilities of the LLM in specific domains, but also improves the efficiency of parameter optimization in the cell manufacturing process. Furthermore, real-time quality monitoring data from each manufacturing step is fed back into the LLM, allowing for intelligent quality assessment and adaptive process optimization. This feedback loop forms the foundation for machine-AI collaboration, driving a self-improving manufacturing system that enhances efficiency, consistency, and product quality. Another focus in the battery manufacturing process is to parse the complex supply chain

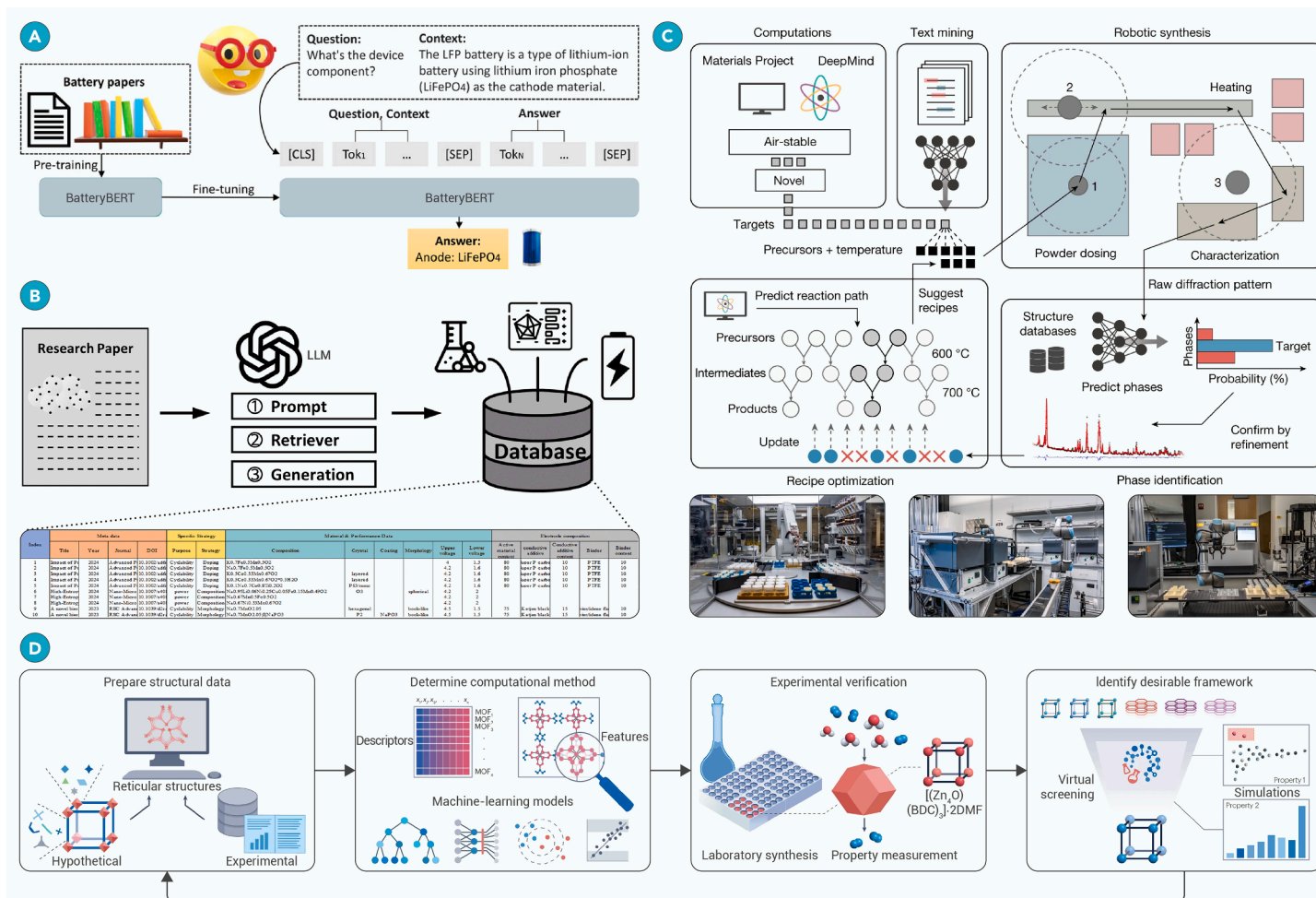


Figure 6. Examples of applications of LLMs in battery materials development (A) A tool for mining information from scientific papers to build a database of battery materials based on an LLM named BatteryBERT. Reproduced with permission.⁴¹ Copyright 2022, American Chemical Society. (B) Examples of autonomous material preparation and testing. Reproduced with permission.⁶⁴ Copyright 2025, Royal Society of Chemistry. (C) Autonomous materials discovery with the A-Lab. Reproduced with permission.⁶⁵ Copyright 2023, Springer Nature. (D) The roles LLMs can take in a data-driven selection process for reticular frameworks. Reproduced with permission.⁶⁶ Copyright 2025, Springer Nature.

relationships from minerals to battery cells, thereby enhancing risk management capabilities. AlMahri et al.¹⁰⁴ proposed a new framework that leverages knowledge graphs and LLM to improve supply chain visibility without relying on direct information sharing from stakeholders, as shown in the Figure 8B. Through a case study of an electric vehicle manufacturer, the framework revealed the dependencies and alternative sourcing options for critical minerals and materials, providing decision-makers with insights into the supply chain structure to support risk management and strategic planning. However, this approach has limitations: the static nature of the knowledge graph cannot reflect the dynamic nature of the supply chain; reliance on publicly available web data may lead to incomplete visibility; and the reliability of the knowledge graph is affected by the credibility of the information sources, with unreliable or outdated data potentially causing inaccuracies.

Additionally, by integrating causal knowledge graphs to strengthen the causal reasoning capabilities of LLMs, they can trace potential causes of quality issues based on textual descriptions of product defects.¹⁰⁵ Intelligent robots driven by LLMs can enhance battery assembly efficiency and product quality.¹⁰⁶ In terms of carbon emission control, LLMs can optimize the entire battery manufacturing workflow, improving energy efficiency and reducing carbon emissions.¹⁰⁷ Moreover, in terms of battery recycling, Huang et al.¹⁰⁸ demonstrated the use of GPT-4 in combination with the green chemical properties of natural polyphenols to optimize the selection of polyphenol types and experimental conditions. Although LLM-based battery manufacturing is still in its early exploratory stage, the development of smart manufacturing technologies and intelligent factory systems, along with the industry's pursuit of cost reduction and efficiency improvement, indicate the significant potential of LLMs in revolutionizing the battery manufacturing sector.

BATTERY SYSTEM MANAGEMENT

Accurate and efficient battery management, such as state estimation and fast-charging strategies, is essential for mitigating battery degradation and enhancing the overall user experience.^{109,110} Traditional approaches to battery health monitoring and safety warnings often rely on manually defined boundary conditions, which can lack adaptability and precision. In contrast, LLMs have recently demonstrated significant potential as innovative tools for intelligent battery management. By summarizing human expertise and learning patterns from diverse data sources, LLMs enable more adaptive and intelligent state estimation and fault detection. Currently, the use of LLMs in battery system management is primarily concentrated in three core areas: state estimation, safety warning, and charge optimization, as illustrated in Figure 9. Currently, research on battery state estimation using LLMs primarily encompasses three main approaches. The first method involves directly fine-tuning a general-purpose LLM with battery sampling data or textually encoding the battery sampling data before fine-tuning.¹¹¹ While straightforward to implement, this approach often yields suboptimal training results. The second method entails training machine learning models or battery mechanism models using battery sampling data first, then integrating these models into the LLM through techniques such as API interfaces or model context protocol.¹⁰³ The LLM decomposes user requirements into task sets, invokes the embedded battery models for simulation calculations, and outputs the results. This approach enables the integration of multiple battery state estimation models into the LLM, effectively compensating for the LLM's limitations in battery simulation capabilities. The third method introduces the LLM into battery machine learning models, leveraging the LLM's reasoning abilities and domain-specific battery knowledge to optimize machine learning algorithms, thereby enhancing the accuracy of battery state

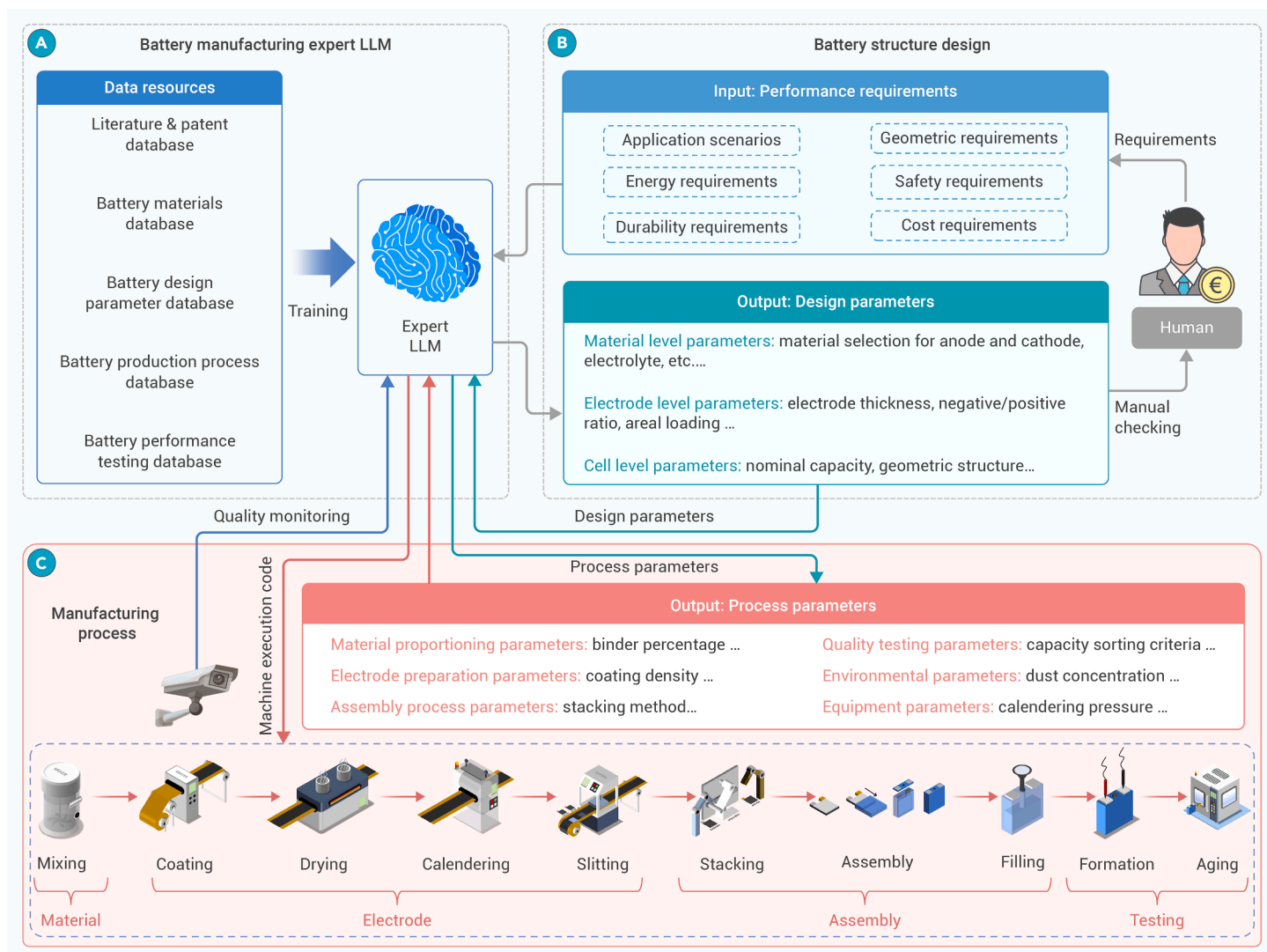


Figure 7. LLMs in intelligent design and manufacturing workflow for batteries (A) Training expert LLMs with battery manufacturing data. (B) Human-LLM collaboration for battery design. (C) LLM-driven automated control of battery manufacturing processes.

estimation.¹¹² This approach is relatively complex, as it requires the development of specialized algorithmic programs.

State estimation

In the field of lithium-ion battery state estimation, precise modeling of state of charge (SOC) and state of health (SOH) is crucial for the safe and efficient operation of battery management systems (BMSs).¹¹³ However, existing mechanistic or machine learning models heavily rely on training data quality, with a key limitation being their overdependence on laboratory-standard databases or accelerated aging data.¹¹⁴ While such data provide benchmark information, they fail to capture the nonlinear impacts of real-world complex factors—such as temperature, charge-discharge rates, dynamic operating conditions, and historical cycling stresses—on battery electrochemical behavior and aging trajectories.¹¹⁴ Neglecting these environmental variables leads models to learn battery characteristics only under idealized conditions. When deployed in practical scenarios like electric vehicles or energy storage systems, their accuracy, robustness, and generalization capability in SOC/SOH estimation significantly deteriorate, ultimately failing to track batteries' true performance boundaries and degradation paths.¹¹⁵

LLM-based battery state estimation technology breaks through this bottleneck with its powerful data-driven capabilities: Its pre-trained models autonomously learn universal deep representations of battery behavior, reducing reliance on manual feature design while mining latent health indicators from massive unlabeled data. The pre-training-fine-tuning paradigm substantially lowers the demand for labeled data, enabling rapid adaptation

to diverse battery types, aging states, and environmental stress patterns (e.g., temperature variations/charge-discharge rates) through few-shot transfer learning, thus bridging the gap between laboratory and real-world applications.¹¹⁶ The self-attention mechanism of LLMs simultaneously captures instantaneous dynamics during individual charge-discharge cycles and long-term aging trends across cycles, while performing end-to-end fusion of multi-source heterogeneous data (time-series electrical signals, environmental parameters, operational condition descriptions). Confronted with inherent defects in real-vehicle data—noise, missing values, and fragmentation—their enhanced fault tolerance and uncertainty quantification capabilities generate more reliable estimations aligned with actual battery performance, providing solid decision-making support for BMSs in dynamic operating environments. Bian et al.¹¹⁷ proposed a hybrid prompt learning approach for SOC estimation of lithium-ion batteries (LIBs) using LLMs. A hard prompt generator is first utilized aiming to convert LIB measurements into instruction text for SOC estimation via language modeling. A soft prompt adapter is then constructed to independently encode a small set of parameter vectors to allow the large language model to efficiently adapt to different estimation tasks while maintaining generalization across LIBs. These vectors are optimized and integrated into the LLM as domain knowledge, enhancing the estimation robustness under different temperatures and operations. Notably, under adverse conditions such as low temperatures or high-load operations, LLMs can rapidly adapt and self-regulate using patterns learned from extensive historical data, providing users with timely and precise feedback on battery status.¹¹⁸

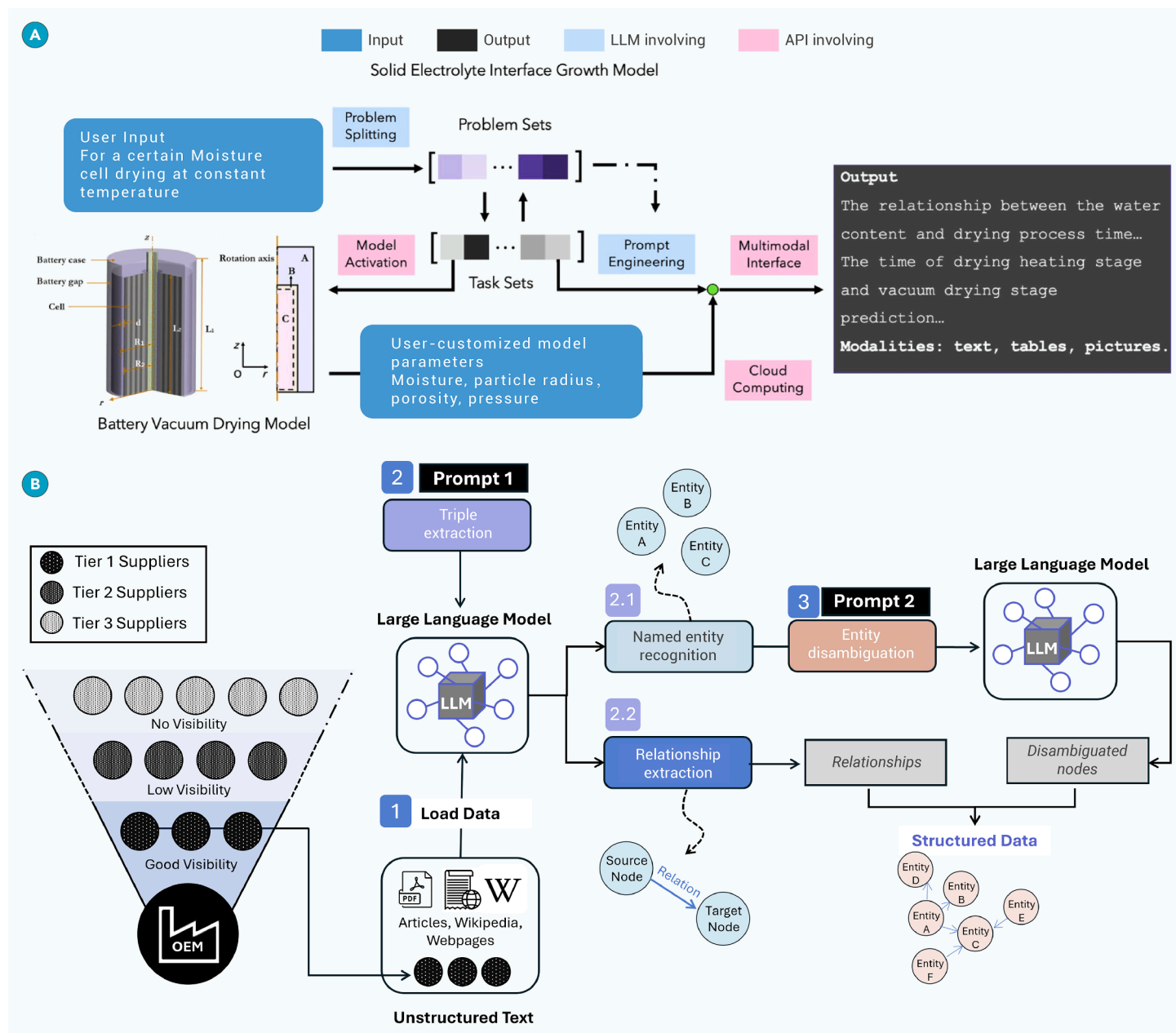


Figure 8. LLM in the battery manufacturing process (A) The detailed workflow of the coevolution between the battery vacuum drying process and LLMs. Reproduced with permission.¹⁰³ Copyright 2025, Elsevier. (B) Diagram of the knowledge graph-large language model framework developed for enhancing supply chain visibility. Reproduced with permission.¹⁰⁴ Copyright 2024, arXiv.

Unlike conventional mechanistic models, LLMs are capable of dynamically adjusting to evolving battery behaviors in the context of large-scale data environments. By analyzing factors such as temperature profiles, charging/discharging cycles, and historical usage data from individual batteries or modules, LLMs can identify underlying degradation patterns and predict future health status.¹¹⁸ For instance, as shown in Figure 10A, Feng et al.¹¹¹ proposed the GPT4Battery framework, which leverages the generalization strength of LLMs to perform cross-battery SOH estimation, significantly reducing the need for extensive monthly degradation experiments. In another study, Lee et al.¹¹⁶ developed a multimodal data-driven platform combining LLMs with automatic graph mining tools to extract battery material data and cycle performance metrics from the literature. This approach enabled the construction of a comprehensive database and the development of machine learning models capable of accurately predicting initial capacity, target cycle performance, and stability in lithium metal batteries. Qiu et al.¹¹⁹ proposed a method for lithium-ion battery state estimation based on prompt-driven fine-tuning. They combined the battery charging and discharging test data with the training text model to construct a specialized dataset for fine-tuning the LLM, thereby accomplishing the battery

state estimation task. Experimental results showed that this method achieved a mean absolute error (MAE) of 0.44% in SOC estimation and a MAE of 0.40% in SOH estimation. Peng et al.¹²⁰ have realized intelligent battery management by integrating internet of things technology and LLMs. They introduced the architecture of an LLM-enabled internet of batteries system, selecting the Lag-Llama probabilistic time-series forecasting base model as the core predictor to effectively forecast the performance of battery system parameters, including future states of charging capacity, discharging capacity, internal impedance, and temperature. Tuncel et al.¹¹² proposed a novel LLM-based technique that has been proposed to automatically implement and optimize machine learning algorithms for battery SOH prediction, as shown in Figure 10B. By integrating ChatGPT into the full machine learning pipeline, the method iteratively improves models through structured prompts, resulting in a 52% reduction in mean absolute percentage error compared with traditional approaches when tested on a public dataset of lithium-ion batteries used in eVTOL aircraft propulsion systems. Li et al.¹⁰³ developed a collaborative evolution system integrating LLMs and physical models, as shown Figure 10C. First, the LLM decomposes user-input questions into relevant problem sets and forms task clusters through

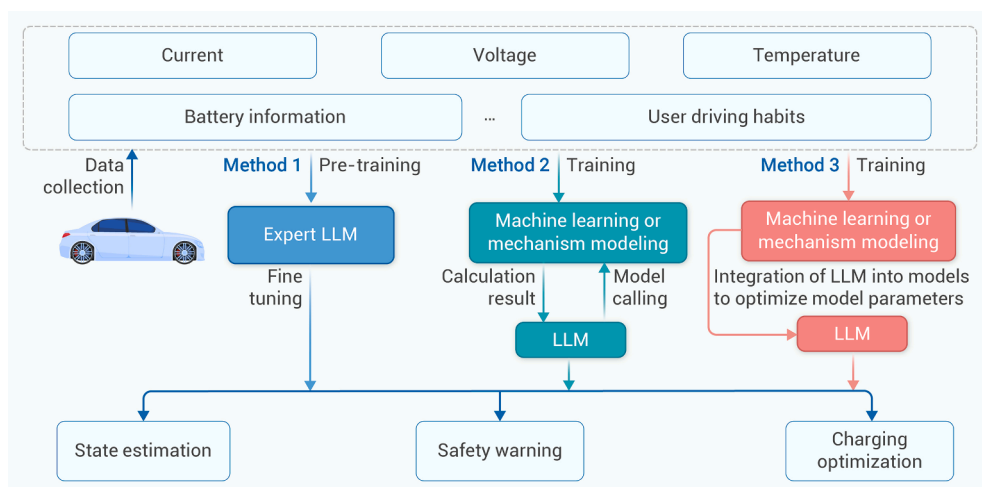


Figure 9. LLM-enabled battery safety warning, state estimation, fast-charging strategy recommendation, and after-sales service

warnings show great promise, they remain in an early stage of exploration. Further research is needed to develop knowledge distillation techniques for lightweight LLM deployment and to address challenges related to real-time, on-vehicle implementation of these intelligent safety systems.

Charging optimization

Optimizing fast-charging strategies is a key challenge in improving the charging efficiency and user experience of electric vehicles. This optimization involves multiple factors,

including charging speed, temperature regulation, and battery lifespan.¹²⁹ To balance user experience with battery lifespan, it is essential to analyze user preferences on their habits, alongside considering the actual state of the battery when formulating a fast-charging strategy.

Safety warning

Leveraging LLM technology, integrating data such as electric vehicle charging prices and regional electricity pricing strategies can reveal how pricing influences user charging decisions.¹³⁰ For example, by analyzing the online behaviors and social media comments of Chinese consumers, it was found that direct current fast charging is the most popular method, especially in first-tier cities.^{131,132} Concerns about finding public charging stations and range anxiety are identified as key issues from the user perspective, underscoring the importance of fast-charging capabilities. Therefore, LLMs can rapidly collect and organize user charging preferences, generating analytical reports that guide the development of fast-charging protocols and the selection of charging equipment. For instance, Namikoshi et al.¹³³ combined LLMs with a multi-armed bandit algorithm to influence consumer preferences for charging methods through personalized dialogue interventions.

Battery failures can lead to performance degradation, operational interruptions, and even serious safety incidents.¹²¹ Therefore, the early detection of faults is critical for minimizing risk and maintaining system integrity.¹²² The user-friendliness and exceptional natural language processing capabilities of LLMs are recognized for their potential to enhance human-machine interaction efficiency in fault diagnosis. Ma et al.¹²³ proposed a novel fault diagnosis pipeline augmented by knowledge graphs. This pipeline employs LLMs to perform complex fault diagnosis tasks while constructing knowledge graphs to improve the models' precision in reasoning. The approach was subsequently evaluated using a proprietary product fault diagnosis database. Experimental results demonstrate that this model outperforms existing RAG models in assisting fault analysis and providing reliable guidance. Compared with traditional threshold-based methods in mechanistic models or fault feature extraction in conventional machine learning algorithms, multimodal LLMs offer a more comprehensive approach. These models can integrate and analyze a wide range of information—including battery sampling data, sensor images, and external environmental conditions—to provide holistic safety assessments.^{124,125} For example, Wang et al.¹²⁶ innovate and validate the temperature excavation (TE) method that interprets the kinetic preferences of thermochemical reactions within minimal experiments into millions of training data. With the help of the TE method, they built the first universally applicable battery thermal runaway model, which achieves high prediction accuracy across a 500°C range on 15 distinct commercial and advanced chemistries with different battery formats and covers all normal working conditions. However, deploying LLM-based safety warning algorithms directly in vehicles poses significant challenges, such as computational overhead and real-time response requirements. To overcome these limitations, Li et al.¹²⁷ developed an LLM agent operating system designed to improve both the performance and efficiency of LLMs in embedded environments. In this system, the LLM functions as the brain, coordinating multiple agents through intelligent resource allocation to enable real-time monitoring and early risk warnings. When deployed on the vehicle side, this agent-based system significantly enhances both response speed and diagnostic accuracy in safety-critical scenarios. Zhang et al.¹²⁸ integrated an LLM with an optimized wavelet packet transform (WPT) to enhance the accuracy of synchronous capacitor fault diagnosis in ultra-high voltage direct current transmission systems. This framework innovatively employs the LLM to automatically optimize WPT parameters, overcoming the limitations of traditional manual parameter selection methods. Compared with conventional approaches, the system maintains high real-time processing efficiency while significantly reducing the false alarm rate. While LLM-based approaches to battery safety

tors, including charging speed, temperature regulation, and battery lifespan.¹²⁹ To balance user experience with battery lifespan, it is essential to analyze user preferences on their habits, alongside considering the actual state of the battery when formulating a fast-charging strategy.

In addition to understanding user behavior, it is crucial to develop fast-charging strategies tailored to specific application scenarios using AI. Traditional fast-charging technologies primarily controlled the charging process by setting fixed power levels and charging times to ensure battery safety and efficiency, while lacking robustness to different battery states, environments, or using habits.¹³⁴ Zhao et al.²² demonstrated the use of LLMs to construct a fast-charging database that enables rapid, accurate information retrieval through user interactions, as shown in Figure 10D. This approach enhances both research efficiency and innovation in fast-charging strategies. Feng et al.¹³⁵ simulates electric vehicle charging behavior by integrating user preferences, psychological traits, and environmental factors to optimize charging decisions.

Beyond power battery state estimation in electric vehicles, LLMs are increasingly deployed in energy storage systems.¹⁸ For energy storage optimization, Gao et al.¹³⁶ demonstrate that the DeepSeek model combines multimodal pre-training with reinforcement learning to establish a spatiotemporal multi-scale intelligent decision system, enhancing the economic efficiency and operational stability of storage stations. Its safety management employs a cross-modal data fusion early-warning mechanism enabling proactive defense across microscopic defects to macroscopic failure pathways. Simultaneously, DeepSeek leverages semantic associations and data-driven modeling to provide quantitative analytical tools for industry chain assessment and strategic planning. This "system symbiosis and energy-efficiency coevolution" paradigm accelerates next-generation energy storage R&D, advancing the development of high-resilience, zero-carbon power systems.

Currently, the application of LLMs in battery intelligent management remains limited. One major challenge is that battery management relies on time-series data, while LLMs, which are typically trained on textual data, do not perform well with time-series information. To advance the use of LLMs in battery intelligent management, it is crucial to develop time-series models, such as TimeGPT,¹³⁷ capable of processing this type of data. Building upon state estimation using temporal large models, it is also necessary to integrate battery text information to achieve state estimation based on multimodal data fusion.

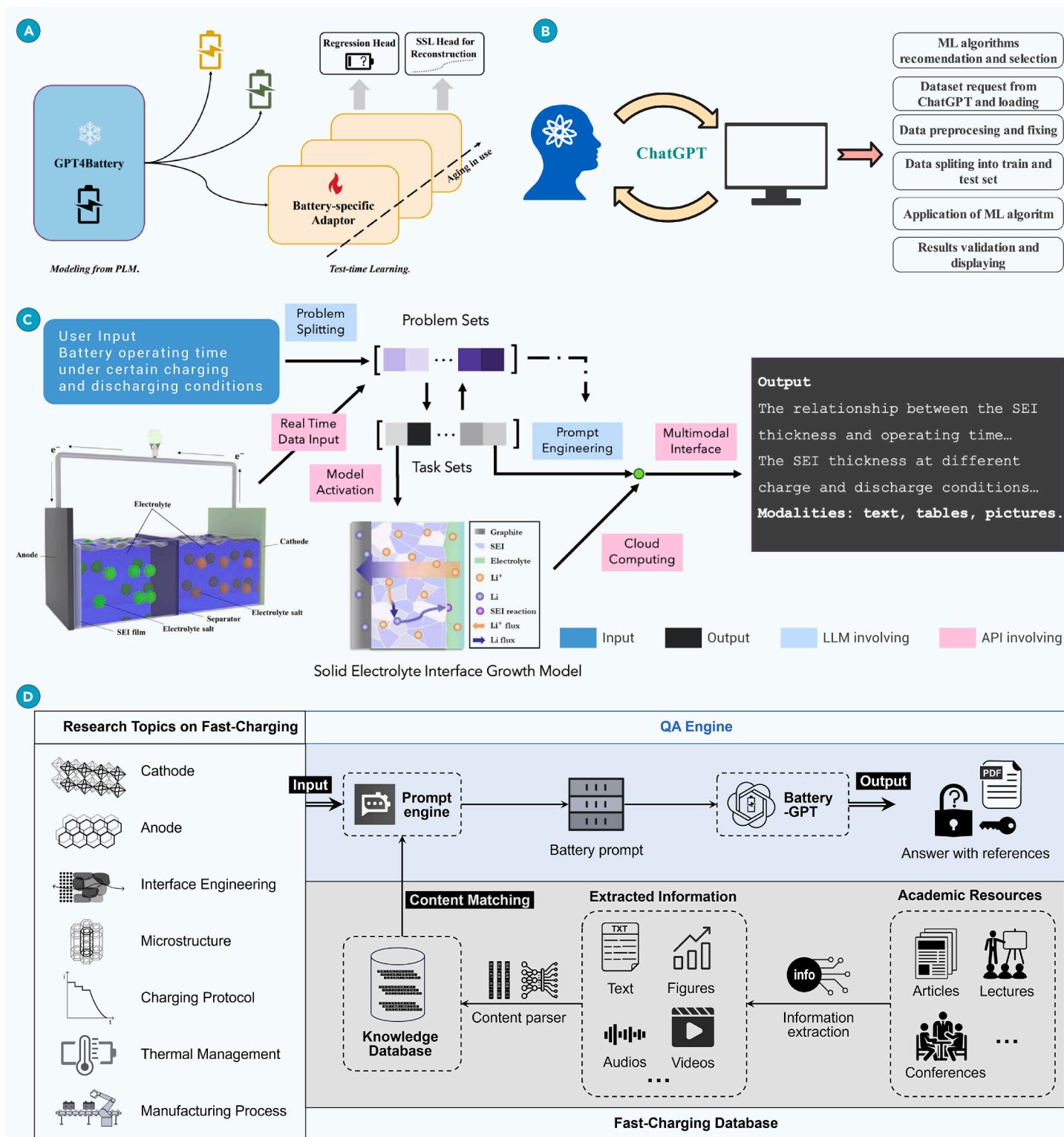


Figure 10. LLMs in battery intelligent management (A) GPT4Battery framework for adaptive SOH estimation. Reproduced with permission.¹¹¹ Copyright 2024, Elsevier. (B) Workflow of the LLM-based methodology for battery state-of-health prediction. Reproduced with permission.¹¹² Copyright 2025, Elsevier. (C) The detailed workflow of the coevolution between the battery vacuum drying process and LLMs. Reproduced with permission.¹⁰³ Copyright 2025, Elsevier. (D) Application of LLMs to an expedited research review for fast-charging technologies. reproduced with permission.²² Copyright 2024, Elsevier.

CHALLENGES AND OUTLOOKS

Challenges

LLM technology has shown promising potential across multiple areas of battery research—including knowledge integration, materials discovery, manufacturing, and intelligent system management. However, significant challenges remain that currently limit its widespread application and real-world deployment.

In the field of battery knowledge integration, LLMs often exhibit limited understanding of highly specialized terminology and domain-specific concepts. This can lead to misinterpretations or inaccuracies in knowledge extraction. While LLMs have demonstrated strong performance in processing textual content from scientific literature, their ability to interpret non-textual data—such as tables, figures, and graphs—remains underdeveloped. This limitation hinders comprehensive information extraction

from full research articles, where critical data are frequently presented in non-text formats.

In terms of battery materials development, LLMs may struggle to accurately comprehend and apply complex physical and chemical principles—particularly at the microscopic or quantum scale. This limitation can lead to generated recommendations or hypotheses that are misaligned with fundamental scientific laws. Additionally, the reliability of LLM-generated content remains largely unverified and requires rigorous experimental validation. Large-scale simulations and data processing tasks involving LLMs demand substantial computational power, posing a barrier for many research institutions with constrained hardware and budgetary resources.

In the domain of battery manufacturing, LLMs can efficiently summarize technical documents, patents, and production guidelines, while also offering insights into process parameters and manufacturing optimization. However, current research still lacks sufficient empirical evidence to validate the feasibility and accuracy of LLM-based process optimization and quality control in industrial environments. Moreover, much of the critical data in battery production is proprietary or confidential, making it difficult to construct comprehensive, high-quality training datasets. As a result, the outputs generated by LLMs may be incomplete or insufficiently precise for practical manufacturing applications.

In battery intelligent management, LLMs face challenges in modeling and interpreting the complex, dynamic electrochemical behaviors of batteries, especially under extreme operating conditions. Their capacity to accurately explain mechanisms such as battery aging or thermal runaway is still limited. Additionally, the development and deployment of high-performing LLMs for battery management require significant investments in hardware infrastructure, data storage, and computational resources—posing further barriers for practical application.

Outlooks

Despite current challenges, LLMs hold tremendous potential to reshape the future of battery research. As their capabilities continue to evolve, LLMs are expected to play an increasingly central role in intelligent, efficient, and scalable solutions across the entire battery research and development landscape.

A key direction for future advancement lies in strengthening the integration and processing of multimodal data. Battery research spans diverse data types—including text, images, tables, and experimental measurements—requiring models capable of consolidating and reasoning across heterogeneous sources. The advent of multimodal general-purpose LLMs has opened new avenues for developing specialized LLMs for battery research. By combining textual content such as experimental reports with microscopic structural images and tabular data, LLMs can achieve a more comprehensive understanding of battery knowledge. Using contrastive learning and crossmodal alignment techniques, LLMs can establish connections between various data formats, including text, images, tabular data, and video inputs. This enhances end-to-end reasoning capabilities, from electron microscopy feature extraction to analytical interpretation. This integrated approach is a promising pathway to accelerate materials discovery, battery design, and battery performance prediction.

To unlock the full potential of LLMs in battery materials development, it is essential to establish closed-loop feedback mechanisms between experimental validation and model refinement. LLM-generated material design proposals should be verified through laboratory experiments, with the resulting data used to iteratively improve model accuracy. An even more ambitious goal is the development of LLM-centered autonomous systems that combine theoretical knowledge with robotic automation to complete the full research cycle—from material proposal generation and screening to synthesis, characterization, and performance testing. To achieving this vision will require close cooperation between academia and industry: enterprises can provide computational power and infrastructure, while academic researchers contribute theoretical models and experimental insights to create an efficient and intelligent discovery loop.

Embedding first-principles knowledge from physics and chemistry into LLMs is crucial for ensuring scientific validity and reliability. The design and performance prediction of battery materials must adhere to fundamental physical and chemical laws—something general-purpose LLMs do not inherently guar-

antee. Integrating domain knowledge into LLM architectures will enhance the accuracy and credibility of model outputs. This process requires advanced AI techniques such as symbolic reasoning over physical equations, bidirectional differentiation in neural network representations, and the integration of structured knowledge graphs. Realizing these capabilities will demand deep interdisciplinary collaboration spanning AI, materials science, electrochemistry, and physics.

In the field of battery manufacturing, enabling secure data sharing while protecting privacy is essential. Manufacturing data are often proprietary and sensitive, limiting its availability for model training. Solutions such as data de-identification, encryption, and federated learning can help overcome this barrier, allowing models to be trained on distributed datasets without compromising confidentiality. These approaches will facilitate the construction of high-quality, diverse public datasets for training domain-specific LLMs, thereby expanding the scope and practicality of AI applications in battery production.

For battery intelligent management, improving the quality and structure of time-series data is critical. Time-series data underpin key battery management tasks such as state estimation, fault detection, and life prediction. Techniques such as data cleaning and augmentation can enhance model robustness, while integrating LLMs with physics-informed models can further improve interpretability and accuracy, especially under complex operational scenarios. Developing customized LLMs specifically tailored for time-series data, such as TimeGPT, represents a promising research direction, combining the strengths of statistical modeling and deep learning to address the unique challenges of battery system monitoring and control.

Finally, fostering cross-disciplinary collaboration and open knowledge exchange is fundamental to accelerating progress. Organizing interdisciplinary workshops, seminars, and collaborative projects can bring together experts from AI, battery science, and related fields to tackle shared challenges. Additionally, establishing open-source communities and shared research platforms will promote the dissemination of new findings, facilitate reproducibility, and support the rapid evolution of LLM applications in the battery domain.

In summary, the deep integration of LLMs into battery research holds vast promise. By advancing multimodal data integration, establishing closed-loop feedback mechanisms, embedding scientific principles, promoting data sharing and privacy protection, improving data quality and model robustness, and fostering cross-disciplinary collaboration and knowledge exchange, the full potential of LLMs in the battery field can be unlocked. These developments will drive transformative progress in battery technology and contribute to the realization of more efficient, reliable, and sustainable energy storage solutions for the future.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.C., Yu Wang, D.G., T.S., X.H., Y.Z., and M.O.; writing – original draft, J.C., Z.L., Yiduo Wang, S.L., W.X., and L.Q.; writing – review & editing, J.C., Yu Wang, and Y.Z.; methodology, J.C., Yu Wang, D.G., T.S., X.H., and Y.Z.; software, J.C., Z.L., Yiduo Wang, S.L., W.X., L.Q., and Y.S.; validation, J.C. and Y.S.; formal analysis, J.C. and Y.S.; data curation, J.C., D.G., and Y.S.; investigation, J.C.; visualization, J.C., Z.L., Yiduo Wang, S.L., W.X., and L.Q.; supervision, Yu Wang, D.G., T.S., X.H., Y.Z., and M.O.; resources, Yu Wang, D.G., T.S., X.H., Y.Z., and M.O.; project administration, Yu Wang, D.G., and Y.Z.; funding acquisition, Yu Wang, X.H., Y.Z., and M.O.; all authors contributed to the manuscript and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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